DEEPAK KATUKURI

AI AUTOMATION & SYSTEMS · LLM WORKFLOWS · FULL-STACK



CONTACT

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TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Core Java, C++

Deep Learning: PyTorch, TensorFlow, CNNs, EfficientNet, neural networks, fine-tuning, timm, Hugging Face

LLM & RAG: Retrieval-Augmented Generation, LangChain, hybrid retrieval (BM25 + dense, RRF), reranking, pgvector, CLIP, DistilBERT, sentence-transformers, contrastive learning, embeddings, Ollama, prompt engineering

Automation & Integration: REST APIs, Webhooks, JSON, workflow automation (LangChain), PL/SQL data-pipeline automation, CI/CD

Full-Stack: React, Vite, Tailwind CSS, React Query, PostgreSQL, FastAPI, Recharts

Machine Learning: Scikit-Learn, classification, ensembles, NumPy, Pandas, Matplotlib, Seaborn

MLOps & Cloud: Docker, Kubernetes, AWS, CI/CD, model deployment

Tools: Git/GitHub, pytest, Jupyter, IBM Watson

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

LANGUAGES

English – Fluent

German – A1/A2 (learning)

PROFILE

AI Automation & Systems engineer and Data Science MSc candidate with 2.5 years of production software engineering and hands-on experience building LLM-driven automation (LangChain, RAG), API-integrated full-stack systems (React + FastAPI + PostgreSQL), and PL/SQL data-pipeline automation. Comfortable across the API/webhook/JSON integration layer and quick to pick up new tooling — most recently validating a hand-rolled retriever against a LangChain rebuild and catching a framework-default regression before trusting it at scale. Seeking an English-speaking working-student role building internal AI automation and systems integrations in Germany.

PROJECTS

RAG Pipeline – Hand-Rolled Retrieval, Validated Against LangChain at Scale

Python · LangChain · BM25 + dense hybrid · pgvector · Ollama · github.com/Deepak589/RAG_pipeline

- Built a hand-rolled hybrid retriever (BM25 + dense bi-encoder, RRF fusion, cross-encoder reranking) from scratch across 4 iterative stages, reaching 0.833 recall@5 / 0.672 MRR on a 55-question labeled golden set before reaching for a framework.
- Rebuilt the same retriever on LangChain over the identical corpus and labels, framework as the only changed variable, and caught a silent 0.06-recall regression from LangChain's default BM25 tokenizer — fixed it to 0.854 recall@5, confirming framework parity before trusting it at scale.
- Scaled the corpus ~100x (7 to ~740 documents, ~3.5GB across arXiv, SEC filings, Wikipedia, and scanned PDFs) with a resumable, file-type-routed ingestion pipeline feeding a pgvector-backed dense store.

RA Nutrition & Lifestyle App – Full-Stack Health Automation

React · FastAPI · PostgreSQL · Recharts · personal product build

- Translated operational health-tracking needs into a shipped full-stack feature: React/Vite/Tailwind frontend over a FastAPI + PostgreSQL backend (4 logical schemas), with JWT auth and token refresh.
- Built a rule-based recommendation engine and a weekly analytics dashboard (Recharts) that closes the loop from raw meal/symptom logs to actionable output for the end user.
- Architected (not yet shipped) an ML forecasting tier — scikit-learn/XGBoost and PyTorch LSTM/GRU/TCN — for flare-risk prediction, scoping the next iteration before building it.

Food Image-to-Recipe Retrieval – Multimodal Deep Learning

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a dual-encoder retrieval model mapping food images (frozen CLIP ViT-B/32) and recipe text (DistilBERT, separate ingredient/instruction streams) into a shared 1024-d space trained with symmetric InfoNCE contrastive loss.
- Reached a validation median rank of 10/200 with semantically correct top-5 results on a 13.5K-image Kaggle dataset, and cut training time ~10x by pre-computing and caching CLIP image features.

EXPERIENCE

Software Engineer
Mar 2022 – Sep 2024

ValueMomentum

- Maintained 2 enterprise P&C insurance and surety-bond applications on OneShield PAS, resolving production defects across policy workflows.
- Migrated full policy, claims, premium, and customer history for the Surety Bonds line from a legacy policy admin system into OneShield PAS, building and validating PL/SQL migration scripts across customer, policy (PID), claims, and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty — the OneShield object copy from the stored object to the loading object silently failed when another bond shared the same bond type under a different bond ID — and kept records current with a weekly backfill script pending a permanent fix.
- Translated business requirements from underwriting and claims stakeholders into technical specs during sprint planning, then accelerated releases by running QA/UAT test cases and supporting CI/CD deployment across the full policy lifecycle.

ML Intern
Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it via IBM Watson ML, communicating results to non-technical reviewers with Matplotlib/Seaborn visualizations.